

TRIAI x Google DeepMind's AI Research foundations course

Challenge 1: Statement of the problem

Project title: Early Detection of Toxic Language through Ubuntu-Aligned Latent Probing of LLMs

Participants of the project

Gebrhans Weldegebriel

Birhanu Simachew

Semere Herruy

Mahendra Pratap

Early Detection of Toxic Language through Ubuntu-Aligned Latent Probing of Large Language Models

Large language models (LLMs) have rapidly become important technologies for a wide range of natural language processing tasks. Their capabilities have enabled their integration into applications such as conversational systems, content generation, and decision-support technologies (Ousidhoum et al., 2021). Despite these advances, the increasing use of Transformer-based LLMs also raises significant concerns about their safety and reliability. Research has demonstrated that language models can generate toxic, offensive, and otherwise harmful content, particularly when they encounter adversarial, provocative, or implicitly biased inputs (Ousidhoum et al., 2021; Luong et al., 2024). Such behavior presents substantial challenges for the responsible deployment of LLMs because harmful outputs may contribute to social damage, reinforce misinformation, and conflict with established safety and ethical requirements, especially when these systems are deployed in publicly accessible or high-stakes contexts (Luong et al., 2024).

Several approaches have been developed to reduce these risks. Contemporary LLMs commonly incorporate safety guardrails, alignment procedures, and external moderation or classification systems designed to identify and prevent harmful outputs. Although these mechanisms can improve the safety of AI systems, they generally provide limited visibility into the internal processes through which a model represents and processes toxic information. In many cases, safety mechanisms function as external or relatively opaque filtering systems, making it difficult to determine where toxic information is encoded within the internal representations of a language model (Waldis et al., 2026). Consequently, an important challenge remains: rather than merely determining whether an LLM produces toxic content, researchers need methods capable of examining the internal representations associated with such behavior.

The effort to understand the internal representations of neural networks predates the emergence of contemporary Transformer architectures. Earlier generations of neural language and vision systems, including convolutional neural networks and recurrent neural networks, generated considerable interest in understanding how information is represented within different layers of neural architectures (Alain & Bengio, 2018). One approach that emerged from this line of research is the use of linear classification probes. Alain and Bengio (2018) demonstrated that

relatively simple classifiers can be used to investigate the information encoded at different intermediate layers of neural networks. Such probing techniques provide a means of examining whether particular information can be recovered from a model's internal representations. Their findings also suggest that information relevant to classification can become increasingly accessible at deeper levels of a neural network.

Although the original work of Alain and Bengio (2018) focused on neural architectures such as ResNet-50 and Inception-v3 rather than contemporary Transformer-based LLMs, the underlying probing approach provides a useful methodological foundation for investigating the internal representations of models such as Gemma. In the context of toxicity detection, latent probing can potentially be used to determine whether information indicating that an input is toxic or safe is already represented within the hidden states of a model before the model produces its final output. This creates an opportunity to move beyond output-level moderation and investigate the representation of toxicity within the model itself.

More recent research has extended this general direction through the development of aligned probing. Waldis et al. (2026) introduced aligned probing as a method for examining relationships between the internal representations of LLMs and their observable behavioral outputs. Their investigation of more than twenty language models, including Llama and Mistral, indicates that LLMs contain information associated with toxicity at both the input and output levels. Their findings further suggest that language models can amplify toxic information beyond what is present in human-generated inputs. These findings make the examination of model internals particularly important because they indicate that toxicity may not simply be a property of generated text; it can also be associated with information encoded within the model's internal states.

The problem is further illustrated by evidence concerning the tendency of pretrained language models to generate harmful language. Gehman et al. (2020) demonstrated that pretrained language models can produce racist, sexist, and other forms of toxic language, creating significant obstacles to their safe deployment. Importantly, toxic behavior does not necessarily require an explicitly toxic prompt; models can sometimes move toward harmful generations even when the initial input does not directly request such content (Gehman et al., 2020). This raises the need for approaches capable of identifying potentially harmful representations before they

are transformed into model outputs. Consequently, examining hidden states may provide an important avenue for detecting toxic information at an earlier stage of the generation process.

There is therefore growing scholarly interest in identifying toxicity within LLMs. Nevertheless, an important limitation remains in the existing approaches. Much of the research on toxicity detection focuses primarily on whether language-model inputs or outputs are toxic, without sufficiently addressing the ethical perspective from which toxicity itself is conceptualized. In particular, there remains a significant gap in methods designed to identify toxic or safe representations within the hidden states of LLMs while explicitly incorporating an African ethical framework.

This gap is particularly important because the development and training of contemporary LLMs are strongly influenced by the data on which they are trained. Many large language models rely heavily on English-language data and datasets originating from technologically developed societies. As a result, the assumptions, values, social relationships, and ethical priorities represented in these systems may not adequately reflect perspectives from African societies. The problem, therefore, is not simply whether an AI system can recognize toxic language, but also which conception of harm, social relations, and acceptable behavior informs that recognition.

This concern connects with a broader philosophical problem concerning whose conception of reason and human value becomes embedded in knowledge-producing systems. Ramose (2005) critically examines the historical assumption that a particular conception of humanity can claim superiority in rationality over other human beings. His critique draws attention to the exclusion of African and other marginalized perspectives from dominant philosophical traditions. Applied to contemporary AI, this philosophical concern raises a related question: whose values and conception of harm are represented in the technologies used to classify and regulate human communication?

A similar concern is raised in contemporary research on AI alignment. Jin et al. (2025) ask whose preferences large language models are ultimately aligned with. This question is equally relevant to toxicity detection. If an AI system is designed to determine whether a particular linguistic expression is harmful, the criteria used to make that determination necessarily reflect some conception of harm and acceptable social behavior. It is therefore necessary to ask whose ethical

perspective is embedded in toxicity-detection systems and whose values are represented in the criteria through which toxic behavior is identified.

From an African philosophical perspective, this issue becomes especially significant when considering Ubuntu. Ubuntu provides an ethical framework that places considerable emphasis on human interconnectedness, community, mutual responsibility, and relational personhood. Rather than treating individuals as completely independent entities, Ubuntu understands human flourishing in relation to other people and the wider community. Agbenonwossi (2025) emphasizes the importance of relational personhood, mutual responsibility, and interdependence within Ubuntu and argues that these considerations are relevant to digital ethics because social harms can affect not only individuals but also relationships and communities.

Within such a framework, harm can be understood not only as an injury to an isolated individual but also as damage to relationships among people and communities. This provides an important ethical basis for reconsidering how AI systems identify toxic language. Hate speech and other forms of harmful communication can damage social relationships, undermine mutual recognition, and contribute to the breakdown of relationships between individuals or groups. Consequently, a toxicity-detection system informed by Ubuntu's value of relationality should not consider harmful language exclusively in terms of its effect on an individual recipient. It should also consider the potential consequences of harmful language for relationships and communal life.

Despite the growing body of research on toxicity detection, probing, and AI alignment, there is a notable absence of toxicity-detection approaches that explicitly incorporate Ubuntu's value of relationality into the investigation of the internal representations of LLMs. Existing approaches provide important methodological foundations for detecting toxic information, but they do not adequately address the question of whether toxicity can be identified within model representations from an African ethical perspective. This represents both a methodological and ethical research gap.

The present study seeks to address this gap by developing a classifier probe informed by Ubuntu's value of relationality. The proposed approach will investigate the hidden representations of a large language model and examine whether information associated with toxic and safe language can be detected within its deeper layers. Rather than treating toxicity

solely as a property of the final generated output, the study will investigate whether potentially toxic information can be identified earlier through the model's internal representations. At the same time, the study will introduce Ubuntu's relational perspective as an ethical lens for understanding toxicity and harm.

The study will focus specifically on the Gemma 1B model. Gemma 1B is part of the Gemma 3 family released in 2025 and is reported to contain 302 million embedding parameters and 698 million non-embedding parameters, with support for a context length of up to 32,000 tokens (Google DeepMind, 2025). By applying latent probing to the internal representations of this model, the research aims to examine whether toxicity-related information can be detected at deeper levels of the model. More importantly, it seeks to explore how such detection can be informed by Ubuntu's conception of relationality.

Accordingly, the central problem addressed by this study is the absence of an approach that combines early toxicity detection within the latent representations of an LLM with an explicitly African ethical framework based on Ubuntu's value of relationality. Addressing this problem can contribute to both the technical study of LLM interpretability and the broader philosophical and ethical discussion concerning whose values should inform AI systems used to evaluate human communication.

References

- Alain, G., and Bengio, Y. (2018). Understanding intermediate layers using linear classifier probes.
- Gehman, S et al. (2020). Real ToxicityPrompts: Evaluating Neural Toxic Degeneration in language models. Allen Institute for Artificial Intelligence.
- Google DeepMind. (2025). Gemma 3 Technical Report.
- Luong, T., et al. (2024). Realistic Evaluation of Toxicity in Large Language Models. Association for Computational Linguistics, pp. 1038-1047.
- Jin, Z et al. (2025). Language Model Alignment in Multilingual Trolley Problems. ICLR, pp. 1-30.
- Ousidhoum, N. et al. (2021). Probing Toxic Content in Large Pre-Trained Language Models. Association of Computational Linguistics, pp. 4262-4274.
- Waldis et al. (2026). Aligned Probing: Relating Toxicity Behaviour and Model Internals. Transactions of the Association of Computational Linguistics, vol. 14, pp. 271-291.